

Functional Analysis: Problems

February 7, 2026

In the following, $F(\mathbb{N}, \mathbb{K})$ denotes the space of all scalar sequences.

1. Suppose $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms on a vector space X , and $S \subseteq X$. Show that S is open with respect to $\|\cdot\|_1$ if and only if S is open with respect to $\|\cdot\|_2$.
2. For $(\alpha_1, \dots, \alpha_n)$ and $(\beta_1, \dots, \beta_n)$ in $\mathbb{K}^n$, and $1 < p < \infty$, show that

(a)

$$\sum_{j=1}^n |\alpha_j \beta_j| \leq \left(\sum_{j=1}^n |\alpha_j|^p \right)^{1/p} \left(\sum_{j=1}^n |\beta_j|^q \right)^{1/q}, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

(b)

$$\left(\sum_{j=1}^n |\alpha_j + \beta_j|^p \right)^{1/p} \leq \left(\sum_{j=1}^n |\alpha_j|^p \right)^{1/p} + \left(\sum_{j=1}^n |\beta_j|^p \right)^{1/p}.$$

3. Let $x, y \in \ell^p$ for $1 < p < \infty$. Using Problem 2(ii), show that $x + y \in \ell^p$ and

$$\left(\sum_{j=1}^{\infty} |x(j) + y(j)|^p \right)^{1/p} \leq \left(\sum_{j=1}^{\infty} |x(j)|^p \right)^{1/p} + \left(\sum_{j=1}^{\infty} |y(j)|^p \right)^{1/p}.$$

4. Suppose $1 \leq r \leq p \leq \infty$. Show the following:

(a) $\ell^r \subseteq \ell^p$,

(b) $\|x\|_p \leq \|x\|_r$ for all $x \in \ell^r$,

(c) $\lim_{p \rightarrow \infty} \|x\|_p = \|x\|_{\infty}$ for every $x \in c_{00}$,

(d) if $r < p$, then ℓ^r is not a closed subset of ℓ^p .

5. For $p > 0$, let

$$X_p := \left\{ x \in F(\mathbb{N}, \mathbb{K}) : \|x\|_p^p := \sum_{n=1}^{\infty} |x(n)|^p < \infty \right\}.$$

Show that if $0 < p < 1$, then there exist $x, y \in X_p$ such that

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

does not hold.

6. Show that if $0 < p < 1$, then the map

$$x \mapsto \left(\int_a^b |x(t)|^p dt \right)^{1/p}$$

does not define a norm on $C[a, b]$.

7. Let X be a normed linear space, and $U, V \subseteq X$. Show that

- (a) if one of U and V is open, then $U + V$ is open,
- (b) if U is compact and V is closed, then $U + V$ is closed.

8. Let X be a linear space and $\nu : X \rightarrow \mathbb{R}$ be a seminorm. Show that

$$Z_\nu = \{x \in X : \nu(x) = 0\}$$

is a closed subspace of X .

9. Let X be a normed linear space and X_0 be a subspace of X . For $x \in X$, define

$$d(x, X_0) = \inf\{\|x - y\| : y \in X_0\}.$$

Prove the following:

- (a) $d(x, X_0) = 0$ if and only if $x \in \overline{X_0}$,
- (b) if X_0 is closed in X and $x \notin X_0$, then $d(x, X_0) > 0$,
- (c) for $x \in X$ and $\alpha \in \mathbb{K}$,

$$d(\alpha x, X_0) = |\alpha|d(x, X_0).$$

10. Let X_0 be a closed subspace of a normed linear space X . Prove the following:

- (a) the map $\nu : x \mapsto \text{dist}(x, X_0)$ defines a seminorm on X ,
- (b) the map $[x] \mapsto \text{dist}(x, X_0)$ defines a norm on the quotient space X/X_0 .

11. The space $P[a, b]$ is not a Banach space with respect to $\|\cdot\|_\infty$.

12. The space c_{00} is not a Banach space with respect to $\|\cdot\|_p$ for any $1 \leq p \leq \infty$.

13. Prove the following:

- (a) $x \mapsto \|x'\|_\infty$ is a seminorm on $C^1[a, b]$,
- (b) $x \mapsto \|x\|_* := \|x\|_\infty + \|x'\|_\infty$ defines a norm on $C^1[a, b]$,
- (c) $C^1[a, b]$ is a Banach space with respect to $\|\cdot\|_*$,
- (d) $C^1[a, b]$ is not a Banach space with respect to $\|\cdot\|_\infty$.

14. Let $X = C[a, b]$ with $\|\cdot\|_1$ and $Y = C[a, b]$ with $\|\cdot\|_\infty$. Show that the identity map

$$A : X \rightarrow Y, \quad Ax = x,$$

is not continuous.

15. Prove the following:

- (a) c_{00} is a proper dense subspace of ℓ^p for $1 \leq p < \infty$, and is not dense in ℓ^∞ ,
- (b) c_{00} is a proper dense subspace of c_0 with respect to $\|\cdot\|_\infty$, and not dense in c with respect to $\|\cdot\|_\infty$,
- (c) $P[a, b]$ is a proper dense subspace of $C[a, b]$ with respect to $\|\cdot\|_p$ for $1 \leq p \leq \infty$.

16. Prove that a Banach space is finite dimensional if and only if every subspace of it is closed.

17. Let X_n , $n \geq 1$, be normed vector spaces. Consider the vector space X of sequences (x_n) such that $x_n \in X_n$ for $n \geq 1$ and

$$\sum_{n=1}^{\infty} \|x_n\| < \infty,$$

endowed with the norm

$$\|(x_n)\| = \sum_{n=1}^{\infty} \|x_n\|, \quad (x_n) \in X.$$

- (a) Prove that if X_n is complete for each $n \geq 1$, then so is X .
- (b) Let X_n^* denote the dual space of X_n , $n \geq 1$. Show that the dual space X^* of X is isometrically isomorphic to the vector space Y of all sequences (f_n) such that $f_n \in X_n^*$ for $n \geq 1$ and

$$\sup_{n \geq 1} \|f_n\| < \infty,$$

endowed with the norm

$$\|(f_n)\| = \sup_{n \geq 1} \|f_n\|, \quad (f_n) \in Y.$$

18. Suppose $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are two norms on a vector space X . Show that the two norms are equivalent if and only if the normed spaces $(X, \|\cdot\|_\alpha)$ and $(X, \|\cdot\|_\beta)$ have the same open sets.